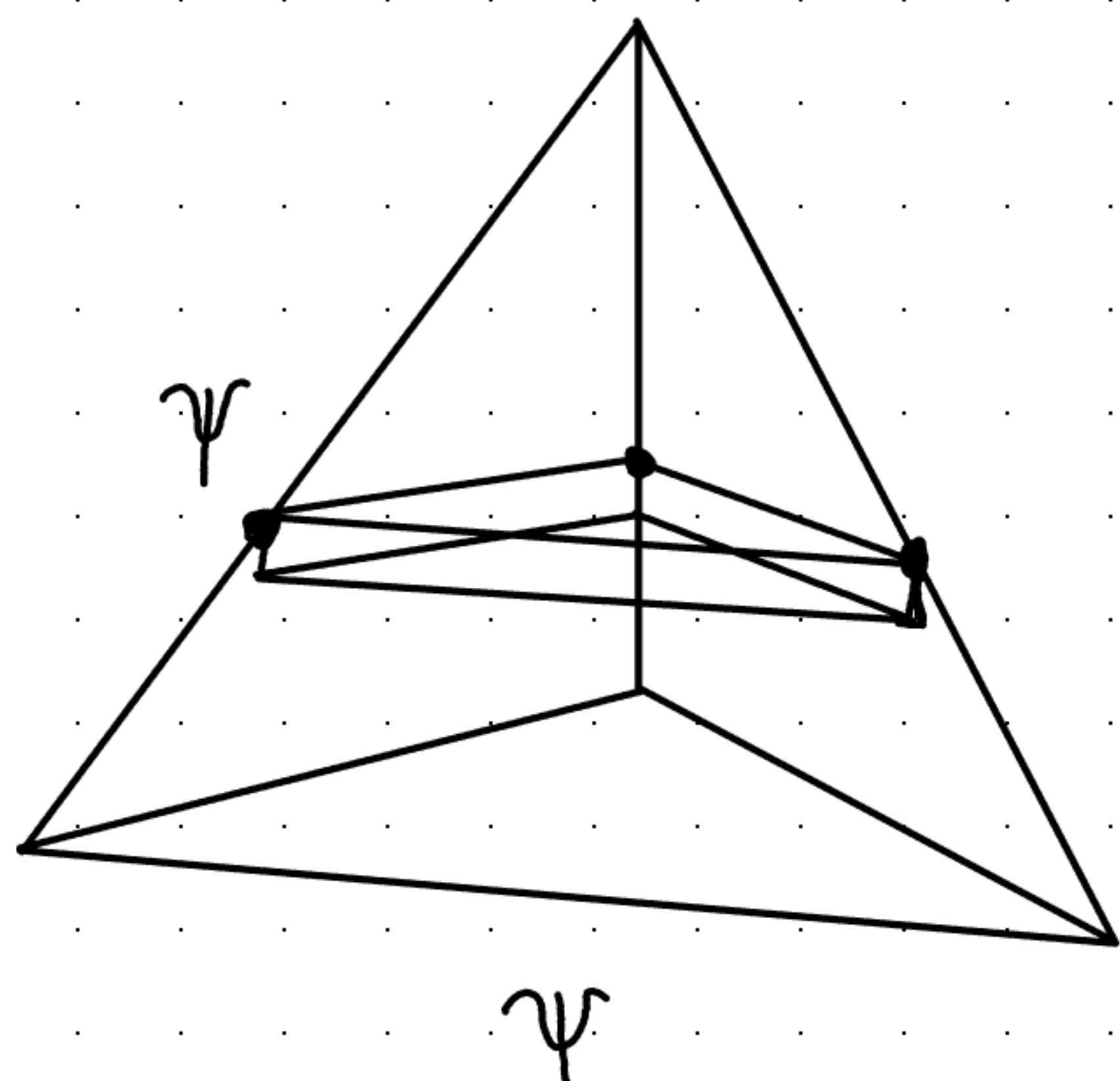
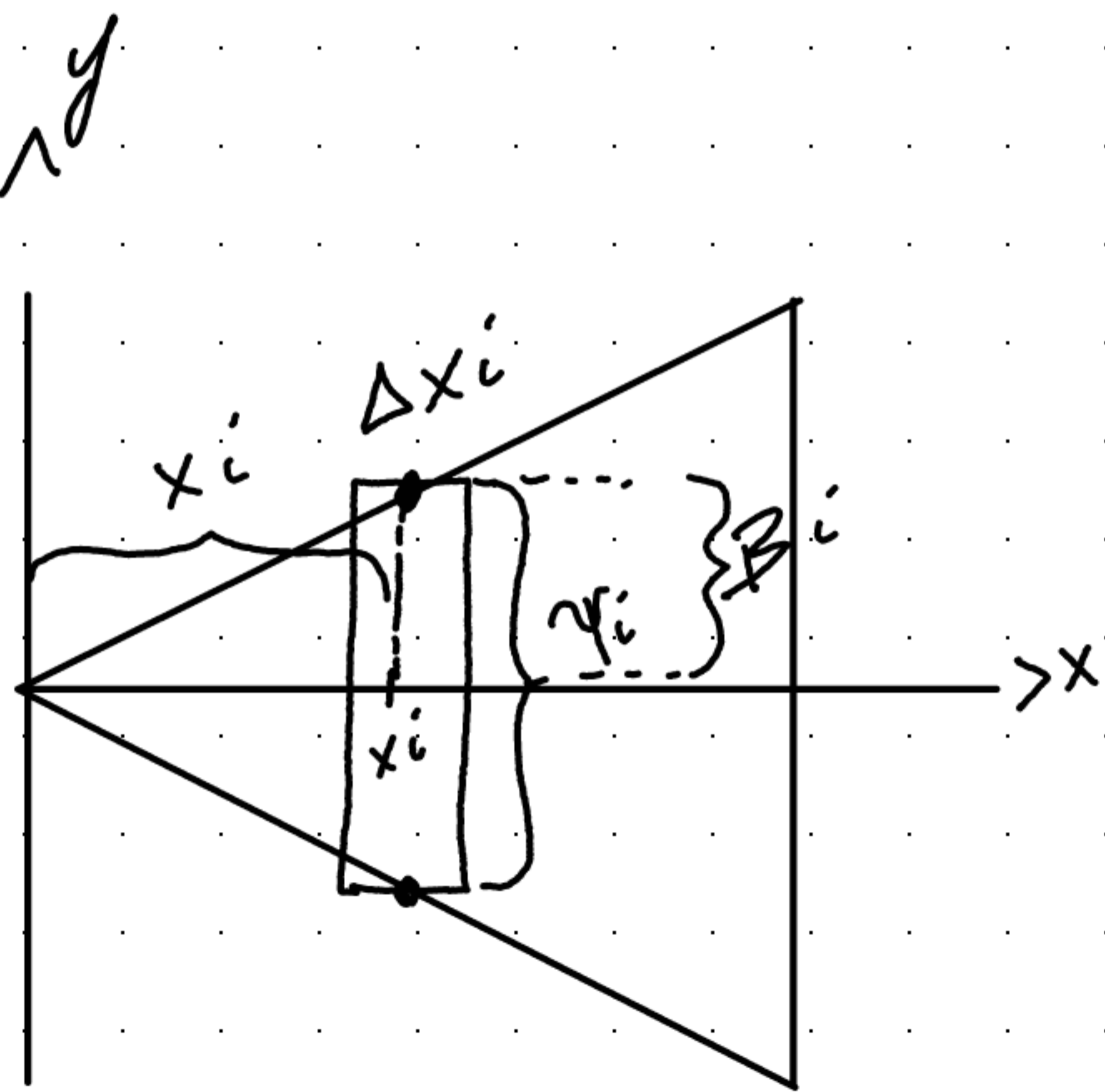


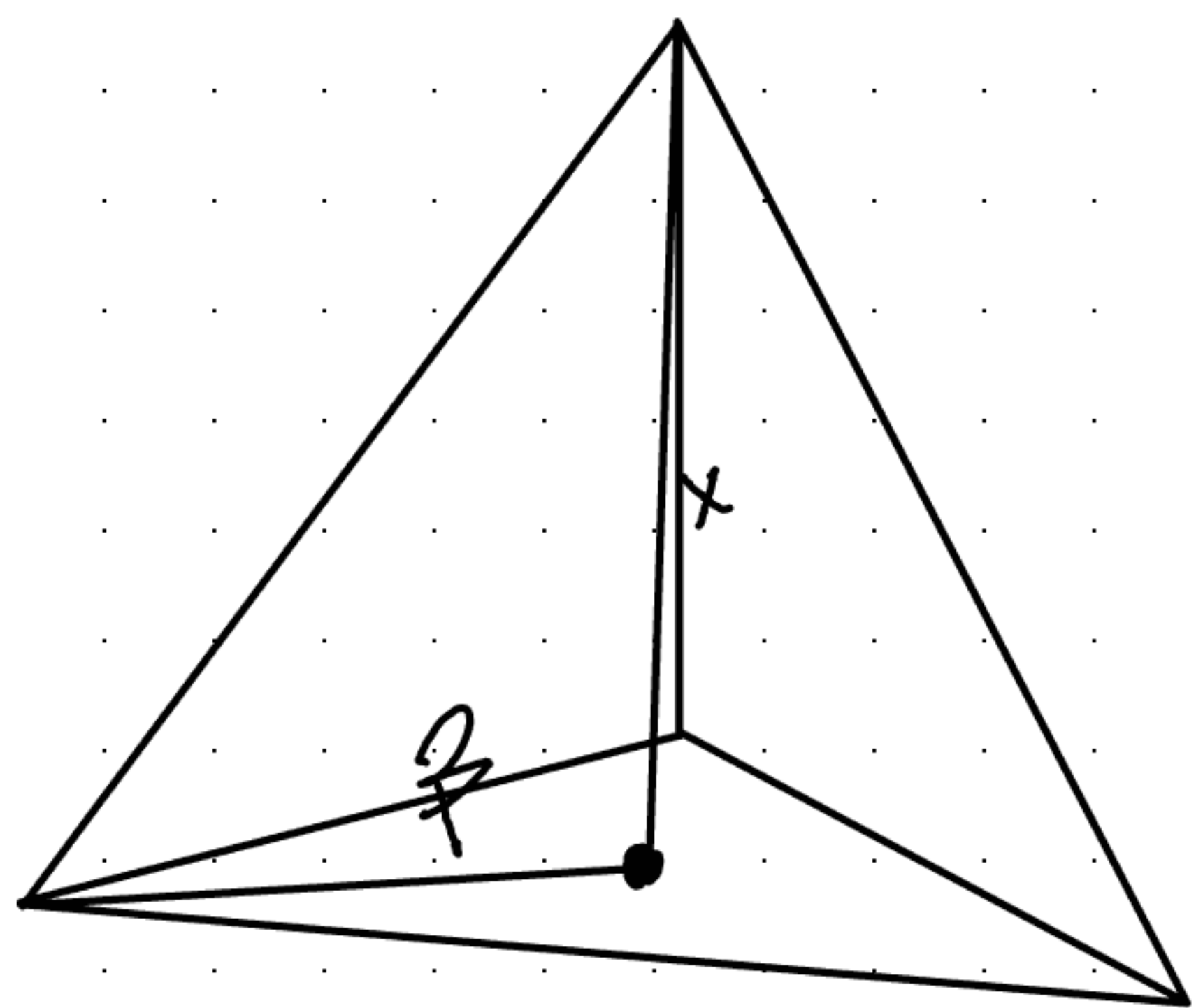
We will use the Slicing method to derive the formula for the Volume of a regular tetrahedron.



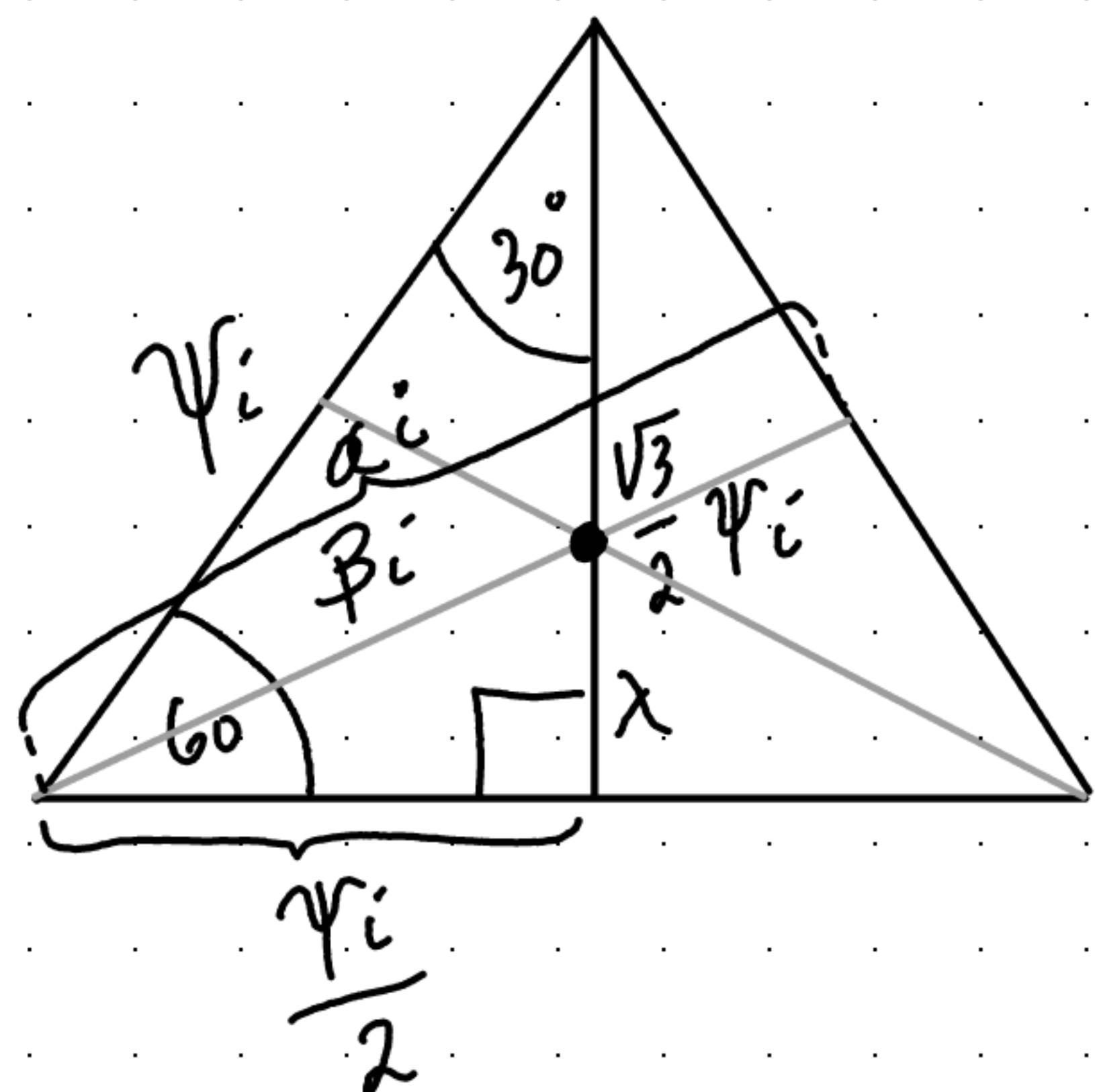
A generalization of the solid whose Volume we are to find with a representative equilateral triangle cross-section drawn in. ψ = the universal side length for all $\psi \in \mathbb{R}$.



We will set the coordinate system so that the solid's apex is at the origin and its altitude spans the x -axis for its entire length. Observe that x_i = altitude of the solid formed by stacking $0-i$ cross-sections.



Visualization of the height (x) element required to enable integration. B = distance from any of the triangle's vertices to the common center for all $B \in \mathbb{R}$.



This is the i^{th} cross-section

ψ_i = common side length of the i^{th} cross-section

α_i = height of the i^{th} cross-section

β_i = distance from vertices to the common center point

$$\frac{\psi_i}{2} = \frac{\sqrt{3}}{2} \beta_i$$

by properties of 30-60-90 triangles

$$\Rightarrow \beta_i = \frac{\sqrt{3}}{3} \psi_i$$

by algebra

$$\lambda_i = \frac{\beta_i}{2}$$

by properties of 30-60-90 triangles

$$\psi_i^2 = \beta_i^2 + x_i^2$$

$$\psi_i^2 = \left(\frac{\sqrt{3}}{3} \psi_i\right)^2 + x_i^2$$

$$x_i = \sqrt{\psi_i^2 - \frac{1}{3} \psi_i^2}$$

$$= \frac{\sqrt{6}}{3} \psi_i$$

Solve for x_i altitude of the solid

obtained by 0-present number

of summations of cross-sections

for all $x \in \mathbb{R}$.

We must now get all variables in terms of x since this is the variable with respect to which we are integrating.

$$\psi_i = \frac{\sqrt{6}}{2} x_i$$

Solve for ψ_i in terms of x_i

$$a_i = \frac{\sqrt{3}}{2} \psi_i$$

Solve for a_i in terms of x_i

$$\begin{aligned} \rightarrow a_i &= \frac{\sqrt{3}}{2} \left(\frac{\sqrt{6}}{2} x_i \right) \\ &= \frac{\sqrt{18}}{4} x_i \end{aligned}$$

$$\begin{aligned} \Delta A_i &= \frac{1}{2} \left(\overbrace{\frac{\sqrt{6}}{2} x_i}^{\text{base}} \right) \left(\overbrace{\frac{\sqrt{18}}{4} x_i}^{\text{height}} \right) \\ &= \frac{\sqrt{108}}{16} x_i^2 \end{aligned}$$

area of the i^{th} cross-section

$\frac{1}{2}$ base $\cdot$ height (of the cross-section)

$$\Delta V_i = \frac{\sqrt{108}}{16} x_i^2 \Delta x_i$$

Volume of the i^{th} cross-section

$$V \approx \frac{\sqrt{108}}{16} \sum_{i=1}^n (x_i)^2 \Delta x_i$$

Riemann approximation of the solid's

Volume

$$V = \lim_{\| \Delta \| \rightarrow 0} \frac{\sqrt{108}}{16} \sum_{i=1}^n (x_i)^2 \Delta x_i$$

limit definition of the Riemann

Sum Volume of the solid

$$V = \frac{\sqrt{108}}{16} \int_0^H (x)^2 dx$$

$$= \frac{\sqrt{108}}{48} [x^3]_0^H$$

$$= \frac{\sqrt{36} \cdot \sqrt{3}}{48} [x^3]_0^H$$

$$= \frac{\sqrt{3}}{8} H^3 \text{ units}^3$$

$$= \frac{\sqrt{3}}{8} \left(\frac{\sqrt{6}}{3} \psi \right)^3$$

$$= \frac{\sqrt{3}}{8} \cdot \frac{2\sqrt{6}}{9} \psi^3$$

$$= \frac{\sqrt{2}}{12} \psi^3 \text{ units}^3$$



Integral representation of the
Solid's Volume

antidifferentiation

algebra

where H = the tetrahedron's
height (x -value) for
an $H \in \mathbb{R}$.

Substitute $x = \frac{\sqrt{6}}{3} \psi$ into
the volume formula for H .

Formula for the Volume of a
tetrahedron where ψ = the
common side length.

Now we will apply our formula for the volume of a regular tetrahedron to Problem 6.2 E #8 from the textbook.

The Problem gives: a tetrahedron with a base side of 4 units.

Our formula for V given the common side length ψ :

$$V = \frac{\sqrt{2}}{12} (\psi)^3 \text{ units}^3$$

Substituting 4 for ψ gives

$$V = \frac{\sqrt{2}}{12} (4)^3 \text{ units}^3$$

$$= \frac{16\sqrt{2}}{3} \text{ units}^3$$